

Unsupervised Learning

Unsupervised Learning is a type of Machine Learning where the training data does not contain predefined labels. Instead of learning the correct answer, the model tries to discover patterns or structures within the data.

A common unsupervised learning technique is clustering.

Clustering groups similar data points together.

Example: A retail company has information about customer purchasing behavior but does not already know what types of customers exist.

A clustering algorithm could identify groups such as:

- Customers who frequently purchase low-cost products
- Customers who make occasional large purchases
- Customers who regularly buy a specific category of product

The company could then use these groups for customer analysis or marketing.

Unlike supervised learning, there is no predefined correct category supplied during training.

Unsupervised learning is useful when organizations have a large amount of data but do not yet know what patterns exist within it.